COMPARISON OF COHOMOLOGY WITH SUPPORT IN THE DERIVED CATEGORY OF MODULES WITH ITS ABELIAN-SHEAF VERSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains a commutative diagram typeset with Xy-pic; the retained chapter preamble supplies the native Xy-pic setup, so it compiles exactly as in the source.

Editorial note. The counted span cites four cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- The distinguished triangle for sections with support. Stacks Project, Cohomology of Sheaves, Lemma lemma-triangle-sections-with-support (tag 0D5L). Provides the triangle in which the comparison morphism is placed.
- Comparison of the derived category of modules with abelian sheaves, and the five lemma. Stacks Project, Cohomology of Sheaves, Lemma lemma-modules-abelian-unbounded (tag 0D5M); Derived Categories, Lemma derived-lemma-five (tag 05SC). Supply the two known isomorphisms and the conclusion from the triangle morphism.

Source reference. Original lemma 0D5Y: sources/stacks-chow/tags.txt, source lines 16503–16503. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0G71: sources/stacks-chow/tags.txt, source lines 20216–20216. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 014A: sources/stacks-chow/tags.txt, source lines 1465–1465. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0G72: sources/stacks-chow/tags.txt, source lines 20217–20217. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $Z \subset X$ be a closed subset. Let K be an object of $D(\mathcal{O}_X)$ and denote K_{ab} its image in $D(\underline{\mathbf{Z}}_X)$.*

1. *There is a canonical map $R\Gamma_Z(X, K) \to R\Gamma_Z(X, K_{ab})$ which is an isomorphism in $D(Ab)$.*
2. *There is a canonical map $R\mathcal{H}_Z(K) \to R\mathcal{H}_Z(K_{ab})$ which is an isomorphism in $D(\underline{\mathbf{Z}}_Z)$.*

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Proof. Proof of (1). The map is constructed as follows. Choose a K-injective complex of $\mathcal{O}_X$-modules $\mathcal{I}^\bullet$ representing K. Choose a quasi-isomorpism $\mathcal{I}^\bullet \to \mathcal{J}^\bullet$ where $\mathcal{J}^\bullet$ is a K-injective complex of abelian groups. Then the map in (1) is given by

$$\Gamma_Z(X, \mathcal{I}^\bullet) \to \Gamma_Z(X, \mathcal{J}^\bullet)$$

determined by the fact that Γ_Z is a functor on abelian sheaves. An easy check shows that the resulting map combined with the canonical maps of Lemma 0D5Y fit into a morphism of distinguished triangles

$$\begin{array}{ccccc} R\Gamma_Z(X, K) & \longrightarrow & R\Gamma(X, K) & \longrightarrow & R\Gamma(U, K) \\ \downarrow & & \downarrow & & \downarrow \\ R\Gamma_Z(X, K_{ab}) & \longrightarrow & R\Gamma(X, K_{ab}) & \longrightarrow & R\Gamma(U, K_{ab}) \end{array}$$

of Lemma 0G71. Since two of the three arrows are isomorphisms by the lemma cited, we conclude by Derived Categories, Lemma 014A.

The proof of (2) is omitted. Hint: use the same argument with Lemma 0G72 for the distinguished triangle. □